

On PADBen, our detectors outperform every reported zero-shot baseline. Note that the IPD task is near-chance for every method including all four PADBen paper baselines, suggesting that this task is universally hard for all the detectors. On RAID, our default single-direction detectors are competitive on chat-tuned generators but weak on base-model (Non-Chat) generators — consistent with our training data containing only chat-tuned model outputs. The task-aware multi-direction extensions substantially close this gap: DetectRL 4-vec reaches 56.3% Overall TPR@5% and 78.9% on Chat generators.

Method	ArXiv		XSum		Writing		Review			
	AUROC	F_1	AUROC	F_1	AUROC	F_1	AUROC	F_1		
Multi-Domain										
Log-Likelihood	65.35	57.55	45.68	41.32	68.00	59.38	75.84	67.22		
Entropy	48.39	29.71	67.84	57.23	39.06	20.55	28.82	2.14		
Rank	57.17	54.62	36.87	22.47	56.26	50.90	55.08	51.90		
Log-Rank	67.01	60.09	46.74	42.60	67.58	57.57	76.40	69.88		
LRR	70.54	61.34	50.09	38.38	64.65	53.09	76.61	68.99		
NPR	53.85	49.65	34.59	18.31	54.96	52.30	50.09	45.39		
DetectGPT	22.15	0.00	12.21	0.00	58.95	50.83	44.43	35.25		
Revise-Detect.	70.40	37.51	50.34	46.07	73.24	64.29	75.01	68.71		
Binoculars	84.03	76.77	77.39	72.18	94.38	79.73	90.00	84.32		
Fast-DetectGPT	43.69	24.46	39.19	28.39	74.21	67.84	77.02	71.62		
RepreGuard [†]	50.00	0.00	50.00	0.00	60.25	60.11	55.15	24.30		
RoBERTa-Base	100.0	100.0	99.99	99.85	99.99	99.65	99.97	99.50		
RoBERTa-Large	99.99	99.90	99.85	98.95	99.54	97.73	99.76	98.90		
XLM-RoBERTa-Base	100.0	100.0	99.97	99.55	99.84	98.76	99.88	99.05		
XLM-RoBERTa-Large	99.98	99.85	99.84	98.95	99.85	98.31	96.40	92.66		
SV-Detect	100.0	100.0	99.94	99.45	99.98	99.60	99.96	99.30		
Method	GPT-3.5		Claude		PaLM-2		Llama-2			
	AUROC	F_1	AUROC	F_1	AUROC	F_1	AUROC	F_1		
Multi-LLM										
Log-Likelihood	62.89	57.80	43.32	28.10	70.03	60.73	75.65	65.90		
Entropy	46.84	23.29	52.25	30.42	45.34	16.56	43.48	66.75		
Rank	52.19	49.32	41.68	22.78	50.40	41.74	57.05	54.40		
Log-Rank	62.84	56.87	43.32	30.12	70.89	63.09	77.97	66.66		
LRR	61.61	52.12	43.30	18.91	71.17	65.51	83.65	75.51		
NPR	50.29	43.81	41.64	32.91	44.64	34.77	52.53	48.68		
DetectGPT	43.46	26.27	32.86	12.56	26.72	0.00	36.71	20.40		
DNA-GPT	61.87	55.04	48.88	25.67	71.48	60.77	75.22	62.89		
Revise-Detect.	70.10	62.72	49.87	27.28	69.84	59.03	75.65	65.87		
Binoculars	88.14	82.50	55.15	39.35	93.30	88.20	96.64	92.30		
Fast-DetectGPT	65.56	59.55	30.01	0.00	65.99	57.58	76.79	69.08		
RepreGuard [†]	66.25	67.53	50.00	0.00	50.00	0.00	50.00	0.00		
RoBERTa-Base	99.97	99.70	99.98	99.80	99.94	99.40	99.84	99.45		
RoBERTa-Large	99.77	98.86	96.23	92.48	97.93	92.64	86.72	76.17		
XLM-RoBERTa-Base	99.88	99.45	98.26	97.48	98.77	97.19	99.69	98.57		
XLM-RoBERTa-Large	99.55	97.56	91.67	84.24	98.73	94.43	99.66	97.67		
SV-Detect	99.99	99.60	99.91	98.91	99.88	99.36	99.99	99.60		
Method	Direct		Prompt		Paraph.		Perturb.		Mixing	
	AUROC	F_1	AUROC	F_1	AUROC	F_1	AUROC	F_1	AUROC	F_1
Multi-Attack										
Log-Likelihood	89.25	82.09	86.87	78.16	64.55	57.59	35.51	0.78	63.70	53.31
Entropy	26.47	0.00	26.18	0.00	48.12	26.01	68.62	68.95	49.37	28.52
Rank	83.50	76.27	81.21	72.86	60.60	52.60	8.04	0.00	52.05	42.46
Log-Rank	89.25	81.45	86.35	77.51	64.69	59.17	37.71	0.78	64.63	56.86
LRR	85.83	77.40	80.80	74.30	63.99	55.20	45.91	29.27	66.12	53.81
NPR	77.98	71.61	77.15	70.63	56.94	46.25	6.78	0.00	48.63	37.65
DetectGPT	52.84	40.90	51.83	37.98	31.79	16.89	18.21	0.00	26.28	0.00
DNA-GPT	88.01	80.78	85.62	77.47	65.61	54.94	40.45	2.73	62.14	50.89
Revise-Detect.	86.88	79.61	84.89	76.21	67.26	62.03	43.98	7.56	65.27	54.39
Binoculars	94.87	89.73	93.45	88.12	88.34	81.56	76.89	69.34	89.12	83.67
Fast-DetectGPT	79.56	72.45	78.43	70.34	70.12	62.89	49.56	41.23	67.23	59.78
RepreGuard [†]	52.78	29.59	50.00	0.00	52.18	26.75	54.81	28.44	54.61	56.41
RoBERTa-Base	99.87	99.60	99.78	99.47	99.67	99.12	98.32	97.45	99.12	98.76
RoBERTa-Large	98.73	97.83	98.45	97.56	97.89	96.78	96.12	94.67	97.56	96.34
XLM-RoBERTa-Base	99.56	99.12	99.23	99.01	98.89	98.34	98.56	97.89	99.01	98.56
XLM-RoBERTa-Large	99.45	98.67	98.89	97.98	98.23	97.67	97.89	96.34	98.67	97.89
SV-Detect	99.99	99.70	99.96	99.56	99.98	99.55	99.99	99.95	99.83	98.75

Table 7: Performance of detectors on DetectRL in the Multi-Domain, Multi-LLM, and Multi-Attack settings. [†]RepreGuard is reproduced under the DetectRL training and evaluation protocol of Sec. 4.1, using the same GPT-Neo-2.7B surrogate as our other rows.

SV-Detect					Fast-DetectGPT				RoBERTa-Base			
Train	ArXiv	XSum	Writing	Review	ArXiv	XSum	Writing	Review	ArXiv	XSum	Writing	Review
Multi-Domain (F_1)												
ArXiv	100.00	93.32	80.78	88.89	24.46	23.71	59.67	60.17	100.0	75.90	77.68	70.69
XSum	93.09	99.45	75.72	88.93	28.43	28.39	62.99	63.08	68.43	99.85	71.79	67.17
Writing	80.57	75.01	99.60	96.64	34.81	33.60	67.84	68.30	78.58	72.72	99.65	94.24
Review	87.93	87.60	97.09	99.30	40.70	37.66	68.25	71.62	82.64	84.15	85.10	99.50

SV-Detect					Fast-DetectGPT				RoBERTa-Base			
Train	GPT-3.5	PaLM-2	Claude	Llama-2	GPT-3.5	PaLM-2	Claude	Llama-2	GPT-3.5	PaLM-2	Claude	Llama-2
Multi-LLM (F_1)												
GPT-3.5	99.60	86.58	90.80	97.36	59.55	59.56	12.96	69.93	99.97	70.34	62.90	94.68
PaLM-2	99.60	98.91	94.83	99.26	55.77	57.58	8.20	68.43	99.25	99.40	93.43	99.25
Claude	99.45	94.87	99.36	98.21	0.19	0.00	0.00	1.18	96.83	83.92	99.80	89.77
Llama-2	99.55	96.41	92.13	99.60	56.28	57.74	8.65	69.08	99.45	93.02	87.56	99.45

SV-Detect					Fast-DetectGPT				RoBERTa-Base			
Train	Prompt	Paraph.	Perturb.	Mixing	Prompt	Paraph.	Perturb.	Mixing	Prompt	Paraph.	Perturb.	Mixing
Multi-Attack (F_1)												
Direct	98.93	96.54	95.46	96.91	64.01	40.45	41.02	31.81	95.73	94.91	64.32	89.07
Prompt	99.56	96.00	91.66	96.45	64.00	39.94	40.40	31.25	97.18	94.98	86.18	92.92
Paraphrase	96.85	99.55	95.89	97.91	61.54	38.32	36.86	27.90	93.66	98.26	78.81	78.81
Perturb	97.50	98.86	99.95	99.10	64.01	40.45	41.14	31.93	87.01	91.46	98.66	91.38
Mixing	99.02	98.46	97.38	98.75	65.89	46.38	45.78	40.93	93.46	91.93	95.26	93.64

Table 8: Cross-source generalization on DetectRL, per-cell F_1 (rows = training source, columns = evaluation source) across the Multi-Domain, Multi-LLM, and Multi-Attack settings.

MIRAGE-DIG (Disjoint-Input Generation)												
Methods	Generate				Polish				Rewrite			
	AUROC	Accuracy	MCC	TPR@5%	AUROC	Accuracy	MCC	TPR@5%	AUROC	Accuracy	MCC	TPR@5%
Likelihood	0.4936	0.5091	0.0183	0.0147	0.4653	0.5000	0.0000	0.0214	0.4337	0.5000	0.0000	0.0148
LogRank	0.4992	0.5128	0.0260	0.0220	0.4512	0.5000	0.0000	0.0195	0.4225	0.5000	0.0000	0.0132
Entropy	0.6522	0.6150	0.2543	0.1099	0.5543	0.5417	0.1247	0.0954	0.5805	0.5566	0.1650	0.1189
RoBERTa-Base	0.5523	0.5397	0.1434	0.1250	0.4859	0.5010	0.0088	0.0460	0.5020	0.5049	0.0293	0.0569
RoBERTa-Large	0.4716	0.5217	0.0842	0.0871	0.5171	0.5151	0.0340	0.0633	0.5570	0.5385	0.0864	0.0895
LRR	0.5215	0.5341	0.0777	0.0701	0.4081	0.5000	0.0000	0.0200	0.3930	0.5000	0.0000	0.0188
DNA-GPT	0.5733	0.5595	0.1196	0.0776	0.4771	0.5004	0.0110	0.0309	0.4453	0.5001	0.0080	0.0251
NPR	0.6120	0.6140	0.2604	0.0191	0.5071	0.5370	0.1071	0.0318	0.4710	0.5201	0.0663	0.0226
DetectGPT	0.6402	0.6258	0.2758	0.0275	0.5469	0.5531	0.1328	0.0355	0.5061	0.5266	0.0826	0.0283
Fast-DetectGPT	0.7768	0.7234	0.4628	0.4310	0.5720	0.5570	0.1293	0.1189	0.5455	0.5432	0.1015	0.1025
ImBD	0.8597	0.7738	0.5497	0.4065	0.7888	0.7148	0.4300	0.2730	0.7825	0.7068	0.4139	0.2933
DetectAnyLLM	0.9525	0.8988	0.7975	0.7770	0.9297	0.8732	0.7487	0.7756	0.9234	0.8705	0.7447	0.7778
SV-Detect (polish-500)	0.9777	0.9241	0.8483	0.8862	0.9113	0.8364	0.6730	0.6414	0.9105	0.8332	0.6672	0.6437
SV-Detect (3-vec task-aware)	0.9912	0.9547	0.9095	0.9579	0.9509	0.8851	0.7706	0.7990	0.9503	0.8856	0.7713	0.7956

MIRAGE-SIG (Shared-Input Generation)												
Methods	Generate				Polish				Rewrite			
	AUROC	Accuracy	MCC	TPR@5%	AUROC	Accuracy	MCC	TPR@5%	AUROC	Accuracy	MCC	TPR@5%
Likelihood	0.4968	0.5207	0.0196	0.0145	0.4599	0.5002	0.0030	0.0233	0.4319	0.5000	0.0000	0.0111
LogRank	0.5008	0.5183	0.0182	0.0186	0.4468	0.5000	0.0000	0.0211	0.4221	0.5000	0.0000	0.0118
Entropy	0.6442	0.6123	0.1592	0.1074	0.5640	0.5439	0.0516	0.0946	0.5858	0.5645	0.0918	0.1198
RoBERTa-Base	0.5368	0.5392	0.0529	0.1101	0.4741	0.5011	0.0048	0.0395	0.5099	0.5122	0.0221	0.0668
RoBERTa-Large	0.4703	0.5236	0.0417	0.0910	0.5150	0.5157	0.0283	0.0702	0.5576	0.5426	0.0405	0.0762
LRR	0.5214	0.5311	0.0314	0.0657	0.4076	0.5000	0.0000	0.0238	0.3978	0.5000	0.0000	0.0174
DNA-GPT	0.5759	0.5647	0.0603	0.0813	0.4788	0.5001	0.0036	0.0340	0.4457	0.5002	0.0048	0.0258
NPR	0.6088	0.6170	0.1571	0.0185	0.5074	0.5277	0.0612	0.0293	0.4738	0.5204	0.0340	0.0177
DetectGPT	0.6353	0.6241	0.1719	0.0193	0.5434	0.5515	0.0668	0.0309	0.5079	0.5260	0.0431	0.0239
Fast-DetectGPT	0.7706	0.7193	0.2078	0.4200	0.5727	0.5619	0.0607	0.1238	0.5480	0.5495	0.0525	0.1097
ImBD	0.8612	0.7791	0.5599	0.4183	0.7951	0.7199	0.4451	0.3036	0.7694	0.6920	0.3936	0.2868
DetectAnyLLM	0.9526	0.9059	0.8119	0.7722	0.9316	0.8740	0.7483	0.7779	0.9158	0.8643	0.7320	0.7574
SV-Detect (polish-500)	0.9779	0.9241	0.8488	0.8839	0.9089	0.8290	0.6583	0.6291	0.9039	0.8265	0.6532	0.6169
SV-Detect (3-vec task-aware)	0.9909	0.9516	0.9032	0.9516	0.9499	0.8837	0.7703	0.7811	0.9421	0.8749	0.7507	0.7706

Table 9: Results across three tasks (GENERATE, POLISH, REWRITE) under two evaluation settings (MIRAGE-DIG and MIRAGE-SIG) on MIRAGE. Metrics reported are AUROC, Balanced Accuracy, MCC, and TPR@5%.